

A Gauge-Free Predictive Model for the Tidal-Sampling Bias of Sun-Synchronous Satellites, Validated on Macrotidal Tidal Flats

Taeyoon Song ^{1,2,*}  and Mirinae Kim ² 

¹ Department of Ocean Sciences, Inha University, Incheon 22212, Republic of Korea

² Haebom Data Inc., #904, Gasan A1 Tower, 205-27 Gasan 1-ro, Geumcheon-gu, Seoul 08503, Republic of Korea; tysong@haebomdata.com (T.S.); mirinae@haebomdata.com (M.K.)

* Correspondence: tysong@haebomdata.com

Abstract: The waterline method for mapping intertidal digital elevation models (DEMs) from optical satellite imagery assumes that scenes sample the local tidal cycle without systematic bias. We derive a closed-form model in which the satellite-sampled tide bias equals $\beta \cdot A \cdot \langle \cos \theta \rangle$, where A is the local tidal amplitude and $\langle \cos \theta \rangle$ the mean cosine of the overpass tide phase, and validate it over the macrotidal Korean coast using 5,082 Landsat-8/9 and Sentinel-2 acquisitions at five tidal flats against hourly tide gauges. Sun-synchronous satellites sample only 60–80 % of the astronomical tidal envelope, missing one extremity. The resulting mean bias has opposite signs on the macrotidal west coast (low-tide bias) and the southern amphidromic coast (high-tide bias). A single slope $\beta = 1.78$ reproduces the bias across five sites and three sensors ($R^2 = 0.98$) and is reproduced from the global FES2022b model alone ($R^2 = 0.983$), enabling a gauge-free correction — though the global model under-estimates the bias magnitude on resonant macrotidal coasts. The bias yields an elevation-domain RMSE of 0.36–1.09 m and unsampled intertidal bands up to 2.5 km wide; because all optical missions cluster near 10:30 local solar time, only phase-orthogonal radar (at a different local solar time) — not higher optical revisit frequency — can close the gap.

Keywords: tidal flat; intertidal DEM; waterline method; tidal aliasing; sun-synchronous orbit; amphidromic system

1. Introduction

1.1. Tidal flats and the need for vertical reference

Tidal flats are among the largest, most biologically productive, and economically important coastal ecosystems on Earth [1,2]. They store carbon, support shellfish economies, host migratory shorebirds, and protect inland areas from storm surges. Mapping their spatial extent and morphology is a precondition for managing coastal change, sea-level rise impact, aquaculture zoning, and ecological monitoring. Because tidal flats are submerged for a large fraction of every day, they cannot be surveyed by conventional optical or LiDAR techniques alone — observations must be coordinated with the tidal stage.

A Digital Elevation Model (DEM) is the most useful single product over tidal flats because it (i) defines the inundation/exposure schedule of every pixel, (ii) supplies the baseline for monitoring vertical change due to deposition, erosion, or sea-level rise, and (iii) supports hydrodynamic modelling. Aerial LiDAR can in principle deliver a centimetre-precision DEM, but coverage is sparse, expensive, and rarely repeated in time. Spaceborne ICESat-2 laser altimetry now provides repeatable, centimetre-level intertidal elevation control over tidal flats, although its sparse, discontinuous ground tracks limit standalone DEM density [3].

1.2. The waterline method

The most widely used satellite-based alternative is the **waterline method** [4–6]. In its simplest form: each cloud-free satellite image taken at acquisition time t delineates a shore-

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line (the land–water boundary) that is, by definition, the iso-elevation contour $z = \eta(t)$, where η is the tide height referenced to a benchmark datum at the nearest gauge. Stacking many waterlines from images at different tidal stages discretises the intertidal hypsometric curve, which is then interpolated into a DEM. Variants of this approach underpin the Digital Earth Australia Intertidal product [7,8], with the sub-pixel accuracy and index sensitivity of the waterline extraction itself characterised by Bishop-Taylor et al. [9], and a growing global literature [10–13]. A complementary line of work derives intertidal elevation more directly, combining ICESat-2 laser altimetry with optical or SAR waterlines to anchor the vertical reference [14,15].

The accuracy of a waterline DEM depends critically on the *sampling of the tidal range* by the available imagery. If imagery is acquired across the full tide cycle with uniform statistical density, the discretisation error is set only by the temporal spacing of the images and by the interpolation method. If, however, satellites systematically miss part of the cycle, the missed elevation band is irrecoverable: no amount of additional imagery acquired at the same time-of-day can fill it.

1.3. Tidal aliasing: recognised qualitatively, not yet predictable

Sun-synchronous orbits place all major Earth-observing optical satellites at near-fixed *local solar times*. The Landsat-8/9 and Sentinel-2 missions all descend across the equator at a local solar time of $\approx 10:00$ – $10:30$, drifting only by a few minutes per year. Because the dominant semi-diurnal tidal constituent (M_2 , period 12.42 h) is not a simple fraction of the solar day, satellite imagery samples the tide at an interval that is incommensurate with the diurnal cycle: each successive overpass at the same site occurs at a slightly different tidal phase, but only in a narrow time-of-day window. Bishop-Taylor et al. [8] documented the resulting *tidal aliasing* for the Digital Earth Australia Intertidal product, introducing spread and low-/high-tide offset indices that show satellite-sampled tide-height distributions to exhibit a non-zero mean bias and a truncated range relative to the underlying tidal cycle; the open-source eo-tides package [16] has since made the diagnostic accessible at global scale. Sent et al. [17] documented an analogous optical-sensor aliasing in the macrotidal Tagus estuary, Portugal, where Sentinel-2 turbidity retrievals are over-represented at spring low tide and under-represented at neap high tide. Despite these recognitions, no closed-form *a priori* model linking the local tide phase at overpass time to a predictable bias amplitude has been proposed.

In the Korean context, the *altimetry-domain* analogue of this phenomenon was reported by Lee, K. et al. [18], who showed that ocean-tide-corrected satellite-altimetry sea-level anomalies near Incheon are sampled preferentially at low tide in the early 1990s and at high tide in 2013–2017, inflating multi-mission sea-level-rise trends by up to approximately $30 \text{ mm} \cdot \text{yr}^{-1}$ before correction. That study, however, addresses the *temporal* drift of the sampling phase within a single altimetry footprint; here we address the *spatial* counterpart for the optical-waterline workflow.

Operationally, the same truncation has been recognised empirically: recent multi-sensor waterline DEM studies on the Korean west coast report that Landsat-8/9 and Sentinel-2 systematically miss the upper intertidal while Sentinel-1 SAR systematically misses the lower intertidal, and that a hybrid optical+SAR collection reaches a stable DEM only after a minimum multi-month observation window [19]. The *physical reason* for that empirical threshold — and for the directional asymmetry of the per-sensor truncation — has not been derived from first principles.

Three open questions remain:

1. **Is the bias systematic and predictable, or noisy and idiosyncratic?**
2. **How does the bias vary across regional amphidromic phase gradients?**
3. **What is the practical translation of the tide-height bias into the DEM elevation domain, and at what scale does it matter?**

Table 1 summarises the position of the present study relative to the existing Korean tidal-flat DEM and sampling-bias literature.

Table 1. Position of the present study within the Korean tidal-flat DEM and altimetry-sampling literature. The present work is distinguished by addressing the optical-waterline sampling bias analytically and *a priori*; prior Korean works map a single-site DEM, or treat the bias empirically or in the altimetry (sea-level) domain.

Study	Region	Sensors	Method	Output	Treatment of sampling bias
Ryu et al. (2002)	Gomso Bay, west coast	Landsat-5 TM	Single-scene waterline	Tidal-flat contour, one epoch	Not addressed
Lee & Ryu (2017)	Ganghwa-do, northwest coast	TanDEM-X (InSAR)	Phase-based DEM	DEM, single mission	Sensor-specific only
Yun et al. (2022)	Ganghwa-do, northwest coast	TanDEM-X (InSAR)	InSAR phase	Operational high-resolution DEM product	Not addressed
Lee, K. et al. (2022)	Incheon, northwest coast	Multi-mission altimetry (T/P-Jason, CryoSat)	Multi-mission tide-corrected trend	Sea-level-rise rate	Temporal sampling-phase bias (altimetry domain)
Lee, J. et al. (2025)	Taeon Peninsula, west coast	L8/9 + S2A/B + S1A	Multi-sensor waterline fusion	DEM, 5-month window optimum versus UAV-LiDAR (MAE 25.6 cm)	Empirically optimised (fuse optical + SAR)
This study	Western + southern coasts (5 sites)	L8/9 + S2 (metadata; 5,082 scenes, 2020–2024)	Phase-bias analysis; $\beta \cdot A \cdot \langle \cos \theta \rangle$	<i>A priori</i> bias-correction model + DEM error budget	Analytically predicted, $R^2 = 0.98$ across sites/sensors

Abbreviations: L8/9 = Landsat-8/9; S2(A/B) = Sentinel-2(A/B); S1A = Sentinel-1A; T/P = TOPEX/Poseidon.

1.4. The Korean macrotidal context

The Korean western coast hosts approximately 2,500 km² of intertidal flats — among the largest in East Asia [20,21] — under one of the world’s largest tidal-range regimes (mean spring range > 8 m at Incheon). On the southern coast (Yeosu, Suncheon Bay), the range falls to about 3 m due to the local amphidromic structure (the regional pattern of tidal-phase rotation around a near-zero-amplitude node), and the phase of high water shifts by about 4 h between the northwest and the south coast of the peninsula [22]. This combination — large amplitude, sharp regional gradient, well-instrumented tide-gauge network — makes Korea an ideal natural laboratory for *quantifying* and *predicting* the satellite tidal-sampling bias.

1.5. Contribution

Where the satellite tidal-sampling bias has so far been treated only diagnostically or empirically [1,7–9,12,19,23–26], this paper makes it *predictable*: we derive and validate a closed-form, gauge-free model that turns a recognised sampling artefact into an *a priori* correction. Specifically, we present an end-to-end analysis that:

- (i) derives a one-parameter analytical model, mean bias = $\beta \cdot A \cdot \langle \cos \theta \rangle$, linking the satellite-sampled tide bias to the local tidal amplitude A and to the mean cosine $\langle \cos \theta \rangle$ of the overpass phase — to our knowledge the first closed-form *a priori* bias-prediction formula for the optical-waterline workflow (Conclusions 1–2);
- (ii) shows that the model explains 98 % of the cross-site, cross-sensor variance and generalises to held-out sites with a root-mean-square error (RMSE) of 0.16 m, and that the slope excess ($\beta = 1.78 \neq 1$) quantitatively decomposes into a spring-neap $\times \cos \theta$ covariance term, providing a physical diagnostic rather than an empirical correction factor (Conclusion 2);
- (iii) provides a quantitative demonstration that the bias *changes sign* across an amphidromic phase gradient — the spatial-domain counterpart to the temporal altimetry-sampling drift previously documented near Incheon by Lee, K. et al. [18] (Conclusion 3);
- (iv) translates the tide-height bias into elevation-domain DEM errors via quantile mapping, yielding quantitative estimates of permanently unsampled intertidal bands (up to 2.5 km wide) and a physically grounded account of why empirically observed

- multi-sensor data-collection-window optima saturate at intermediate timescales (Section 5.3; Conclusions 4–6);
- (v) reproduces the fit from a global ocean tide model alone (FES2022b: $R^2 = 0.983$ in a self-consistency check that supplies both the overpass-time tides and the comparison reference), indicating that the *a priori* correction can be computed without local tide-gauge data (Conclusion 7).

The Korean macrotidal coast (five sites, 2020–2024; Section 2) provides a natural laboratory for the validation, because it spans both a strongly negative-bias regime (the macrotidal west coast) and a sign-reversed positive-bias regime (the southern amphidromic regime) within a single, densely instrumented tide-gauge network. Although the validation is Korean, the model itself is site-independent: because every operational optical Earth-observation mission shares the same near-10:30 sun-synchronous overpass, the predicted bias and its gauge-free correction are expected to transfer to other macrotidal coasts (Section 4.8).

2. Study area and data

2.1. Sites

Five tidal flats along the western and southern Korean coast were chosen to span the full range of local tidal amplitudes and regional phase offsets (Figure 1; Table 2):

- **Ganghwa-do** (mean spring range, MSR, ≈ 8 m): the largest single tidal-flat complex in Korea, exposed for approximately 50 % of the tidal cycle.
- **Garorim Bay** (MSR ≈ 6 m): semi-enclosed bay, fringed with tidal flats.
- **Gomso Bay** (MSR ≈ 6 m): cusped tidal flat with strong fluvial input.
- **Hampyeong Bay** (MSR ≈ 4 m): bay-mouth tidal flat at the transition to the south coast.
- **Suncheon Bay** (MSR ≈ 3 m): reed-dominated estuary, Ramsar-listed wetland.

Each site is matched with the nearest principal tide-gauge of the Korea Hydrographic and Oceanographic Agency (KHOA; Figure 1, triangles).

Garorim Bay lies approximately 30 km north of the Taean Peninsula sub-region addressed by recent multi-sensor west-coast DEM studies [19] but does not overlap with them; the present analysis therefore provides an independent measurement of the macrotidal west-coast sampling geometry adjacent to, rather than within, that area. Suncheon Bay further extends the coverage to a south-coast amphidromic regime not previously sampled in this literature.

2.2. Tide-gauge reference: KHOA Open API

Hourly quality-controlled tide observations for 2020-01-01 to 2024-12-31 were retrieved through the Korea Hydrographic and Oceanographic Agency (KHOA) Open API, hosted on the Korean public data portal (apis.data.go.kr/1192136/hourlyTide). For each of the five gauge stations we accumulated 42,674–43,839 hourly samples (> 99.5 % coverage). Tide heights are stored in the *KHOA datum* (approximate lowest low water, ALLW — the sum of the amplitudes of the four principal constituents M_2 , S_2 , K_1 , O_1 below mean sea level, which is distinct from the lowest astronomical tide); we report all elevations in this datum. Brief acquisition outages (≤ 0.3 % of dates) and four anomalous days at Incheon (DT_0001; 2020-10-01 to 2020-10-04) are tolerated as random missing data.

2.3. Satellite metadata: Google Earth Engine

Per-scene metadata (acquisition time, scene ID, cloud-cover fraction, WRS/MGRS tile) were extracted from Google Earth Engine [27] for three optical sensors over the 2020–2024 period: Landsat 8 (LANDSAT/LC08/C02/T1_L2; 2013-04–present), Landsat 9 (LANDSAT/LC09/C02/T1_L2; 2021-10–present), and the European Sentinel-2 mission (COPERNICUS/S2_HARMONIZED; 2015-06–present; [28]).

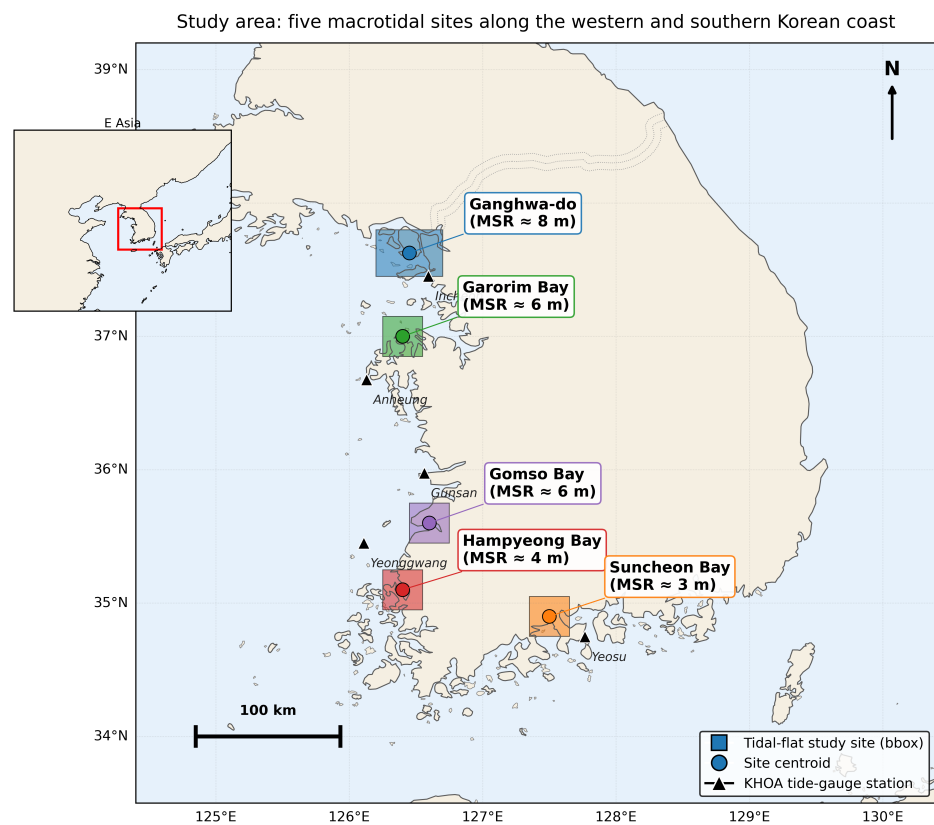


Figure 1. Study sites along the western and southern Korean coast. Coloured boxes show the satellite-acquisition bounding boxes; circles mark site centroids; triangles mark KHOA tide-gauge stations (Incheon (DT_0001), Anheung (DT_0067), Gusan (DT_0018), Yeonggwang (DT_0003), Yeosu (DT_0016)). MSR = mean spring range (literature estimate). The five sites span a mean spring tidal range from approximately 8 m (Ganghwa-do) to about 3 m (Suncheon Bay) and straddle the M_2 amphidromic phase gradient of the Yellow Sea — the combination that drives the sign-reversing sampling bias. Inset: regional context.

Pixel data were not required: this is a metadata analysis. Scenes intersecting each site's bounding box were retained, with a permissive cloud-cover threshold $\leq 60\%$; the cloud-screened sample sizes ranged from 500 (Gomso) to 1,876 scenes (Garorim Bay; Table 2).

Table 2. Data inventory per site for 2020–2024. Planar slopes are the literature-based first-order values used for the horizontal-domain error budget (Section 3.5); see that section for sources.

Site	KHOA gauge	Hourly records (n)	Sat. scenes (L8 / L9 / S2)	Planar slope (m/km)
Ganghwa-do	Incheon (DT_0001)	43,689	137 / 90 / 826	0.8
Garorim Bay	Anheung (DT_0067)	43,720	129 / 91 / 1,656	1.5
Gomso Bay	Gunsan (DT_0018)	42,674	203 / 110 / 187	1.2
Hampyeong Bay	Yeonggwang (DT_0003)	43,839	195 / 119 / 796	1.5
Suncheon Bay	Yeosu (DT_0016)	43,688	70 / 38 / 435	2.0

Tide heights at each satellite acquisition time were obtained by linear interpolation of the hourly KHOA series between bracketing hours.

3. Materials and Methods

3.1. Aliasing metrics

For each (site, sensor) combination, we use the standard set of aliasing metrics introduced for sun-synchronous waterline analysis [8,16]:

- **Spread** = $(\max \eta_{\text{sat}} - \min \eta_{\text{sat}}) / (Q_{\text{ref}}(0.999) - Q_{\text{ref}}(0.001))$. Fraction of the reference tidal range covered by satellite samples.
- **Low offset** = $\max(0, \min \eta_{\text{sat}} - Q_{\text{ref}}(0.001)) / \text{range}$. Fraction of the reference range below the lowest satellite sample.
- **High offset** = $\max(0, Q_{\text{ref}}(0.999) - \max \eta_{\text{sat}}) / \text{range}$. Fraction above the highest satellite sample.
- **Kolmogorov–Smirnov (KS) statistic** between satellite-sampled and reference tide-height empirical cumulative distribution functions (CDFs).
- **Mean bias** = $\langle \eta_{\text{sat}} \rangle - \langle \eta_{\text{ref}} \rangle$, in metres (KHOA datum).

3.2. Tidal phase

Local high-water (HW) events were extracted from the hourly KHOA series by a peak-finding algorithm with a minimum inter-peak separation of 8 h (implementation: SciPy). For every satellite acquisition time t we then computed the **normalised phase** since the bracketing prior HW:

$$\phi(t) = \frac{t - t_{\text{HW,prev}}}{t_{\text{HW,next}} - t_{\text{HW,prev}}} \in [0, 1).$$

We associate ϕ with an angle $\theta = 2\pi\phi$ such that $\theta = 0$ corresponds to HW and $\theta \approx \pi$ to low water (LW). For each (site, sensor) we compute circular statistics of $\{\theta_i\}$:

- Mean resultant vector $\langle \cos \theta \rangle, \langle \sin \theta \rangle$,
- Mean resultant length $R = \sqrt{\langle \cos \theta \rangle^2 + \langle \sin \theta \rangle^2} \in [0, 1]$,
- Circular mean phase $\bar{\theta}$ and circular standard deviation.

3.3. Analytical bias model

For a near-symmetric, M_2 -dominated tide, the elevation at time t can be written

$$\eta(t) \approx A_i \cos(\theta_i) + \eta_0(t),$$

where A_t is the slowly varying envelope and η_0 the mean water level. The bias inherited by sampling at the satellite times $\{t_i\}$ relative to the dense reference is, to first order,

$$\text{bias} = \langle \eta_{\text{sat}} \rangle - \langle \eta_{\text{ref}} \rangle \approx \langle A \cos \theta \rangle.$$

Writing $A = \langle A \rangle + A'$ and $\cos \theta = \langle \cos \theta \rangle + (\cos \theta)'$, this expands as

$$\langle A \cos \theta \rangle = \langle A \rangle \cdot \langle \cos \theta \rangle + \text{cov}(A, \cos \theta),$$

so that the bias can be written

$$\text{bias} \approx \beta \cdot A \cdot \langle \cos \theta \rangle, \quad (1)$$

with $A \equiv \langle A \rangle$ the time-mean amplitude and an effective proportionality factor

$$\beta = 1 + \frac{\text{cov}(A, \cos \theta)}{\langle A \rangle \cdot \langle \cos \theta \rangle}.$$

The leading-order prediction is $\beta = 1$ when A and $\cos \theta$ are uncorrelated. Any non-zero covariance — for instance, when spring tides (large A) systematically coincide with a preferred sign of $\cos \theta$ at the satellite overpass time — shifts β away from unity. The empirical departure $\beta > 1$ documented in Sections 4 and 5 is therefore itself a *diagnostic* of physical mechanisms that produce $\text{cov}(A, \cos \theta) \neq 0$. We fit β by ordinary least squares (OLS) on the 15 (site, sensor) pairs.

3.4. Stability tests

We test the temporal/sensor robustness of Eq. (1) by partitioning the data along four axes and re-fitting:

- **Annual:** 5 fits, one per calendar year 2020–2024.
- **Seasonal:** 4 fits, one per DJF / MAM / JJA / SON.
- **Sensor:** 3 fits, one per L8 / L9 / S2.
- **Leave-one-site-out:** 5 fits, each training on 4 sites (12 points) and predicting the 5th site's 3 sensors.

A 2,000-resample bootstrap of the pooled ($n = 15$) fit yields confidence intervals on β and on the intercept.

3.5. Conversion to elevation-domain DEM error

Because every waterline is mapped to its tide-stage elevation, the bias in the η -distribution maps directly into the elevation domain. For each cumulative probability $p \in [0.005, 0.995]$ we compute

$$z_{\text{true}}(p) = Q_{\text{ref}}(p), \quad z_{\text{sat}}(p) = Q_{\text{sat}}(p), \quad \varepsilon(p) = z_{\text{sat}}(p) - z_{\text{true}}(p).$$

Aggregate elevation-domain metrics:

- **Vertical DEM RMSE** = $\sqrt{\langle \varepsilon(p)^2 \rangle}$,
- **Mean elevation bias** = $\langle \varepsilon(p) \rangle$ (= η -domain mean bias),
- **Truncated low / high bands** = $\max(0, \min \eta_{\text{sat}} - Q_{\text{ref}}(p_{\text{min}}))$ and $\max(0, Q_{\text{ref}}(p_{\text{max}}) - \max \eta_{\text{sat}})$; these are intertidal elevation bands that satellites never sample.

Assuming a locally planar tidal flat with slope s , vertical errors are converted to **horizontal contour displacement** as $\Delta x = \Delta z / s$. Site-specific slopes were assigned from published Korean tidal-flat geomorphology surveys and DEM products: Ganghwa-do = 0.8 m/km [25,26], Garorim Bay = 1.5 m/km [19], Gomso Bay = 1.2 m/km [23,24], Hampyeong Bay = 1.5 m/km (assumed comparable to Gomso based on [20]), and Suncheon Bay = 2.0 m/km (assumed from south-coast steeper-flat regimes; literature constraint weaker).

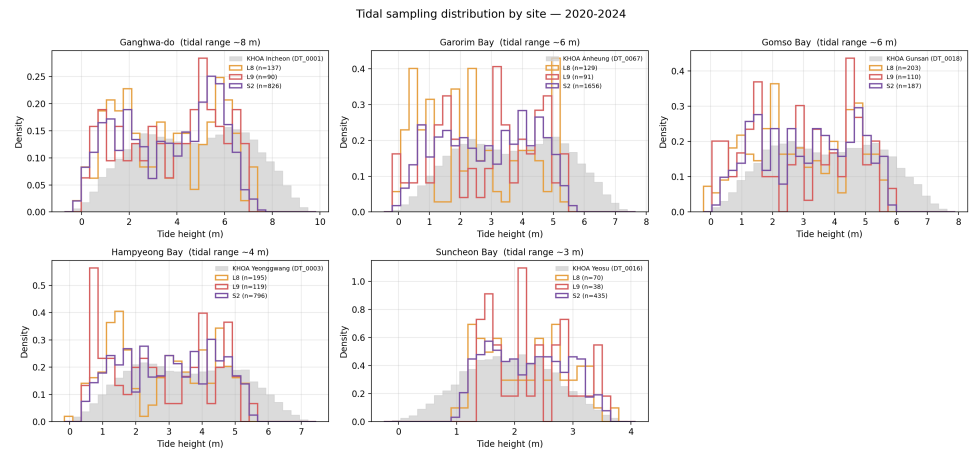


Figure 2. Empirical tide-height distributions at satellite acquisition times (coloured outlines) versus the 5-year KHOA reference (grey filled). Each panel is a site; colours mark sensors (L8 = orange, L9 = red, S2 = purple). The mirror-image pattern between Suncheon Bay (under-samples low tide) and the four western sites (under-sample high tide) is the visual signature of the sign-reversing bias quantified throughout Section 4.

These are first-order values intended to give an order-of-magnitude horizontal-error budget; a refined analysis with site-specific LiDAR or SRTM hypsometric curves would adjust the horizontal — but not the vertical — numbers proportionally.

3.6. Implementation

All analyses are reproducible from open data through the GitHub repository released with this paper (Data and code availability). The processing pipeline uses GEE-API metadata extraction (no pixel downloads), KHOA Open API caching, pandas/NumPy analysis, SciPy for linear regression and bootstrap, and matplotlib/Cartopy for figures. Throughout the paper, the reference distribution against which satellite sampling is compared is the KHOA observed hourly series (Section 2.2). Two complementary astronomical-only references are used as sensitivity tests in Section 4.7: (i) harmonic-tide reconstruction of the same KHOA records (UTide; [29]), which removes weather-induced residuals but preserves site-specific constituent amplitudes derived from local observations; and (ii) the global ocean tide model **FES2022b** ([30]; AVISO ocean_tide_extrapolated grids at $1/30^\circ$ resolution), from which per-constituent amplitudes and phases for eight major constituents (M_2 , S_2 , K_1 , O_1 , N_2 , P_1 , K_2 , Q_1) are interpolated at each KHOA gauge coordinate and harmonically synthesised at hourly cadence over 2020–2024. Because FES2022b supplies both the overpass-time tides and the comparison reference, this variant is an internal self-consistency check on a globally available model (the same model on both sides of the regression) rather than a test against independent ground truth; the only ground-truth comparison is FES-predicted versus KHOA-observed bias (Section 4.7), which matches in sign but under-estimates magnitude.

4. Results

4.1. Cross-site tidal sampling distributions reveal a sign-reversing bias

Figure 2 shows the empirical distribution of tide heights (in the KHOA datum) at satellite-acquisition times against the 5-year KHOA reference for each of the 5 sites, broken out by sensor. At every site, satellite distributions are visibly non-uniform relative to the reference. At the western sites (Ganghwa, Garorim, Gomso, Hampyeong), satellite samples are over-represented in the lower half of the elevation range and empty above 6–7 m KHOA datum (approximately the top 20 % of the histogram). At Suncheon Bay, the pattern is opposite: satellites over-sample the upper half of the range and miss the lowest 0.3–1 m. Figure S1 (CDF panels) reinforces the same finding in cumulative form.

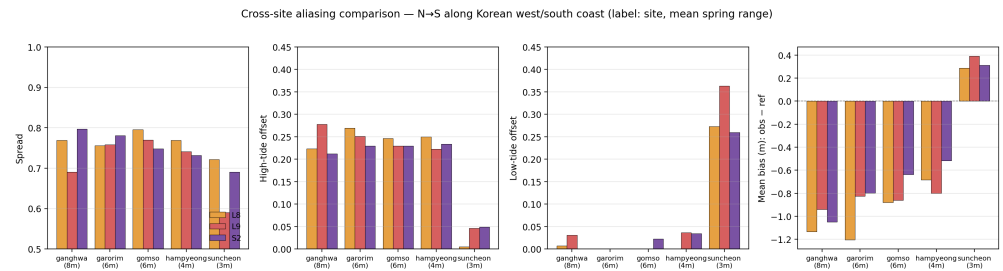


Figure 3. Cross-site aliasing metrics by sensor. Columns: spread, high-tide offset, low-tide offset, and mean bias (obs – ref). Bars: L8 (orange), L9 (red), S2 (purple). Sites are ordered N→S along the western/southern coast; horizontal label gives the literature mean spring range. The sign-reversal of mean bias and the swap between high- and low-tide offsets at Suncheon Bay relative to the western sites — observed consistently across all three sensors — provide quantitative confirmation that the bias polarity is set by the regional amphidromic geometry, not by the sensor or the tidal amplitude alone.

Quantitatively, the aliasing metrics (Section 3.1) confirm the sign reversal (Figure 3, Table 3). The four western sites all exhibit a high-tide offset of 21–28 % and a low offset near 0 %; Suncheon Bay alone shows a low-tide offset of 26–36 % with a high-tide offset of 0–5 %. Mean tide-height bias is –0.5 to –1.2 m at the western sites and +0.3 m at Suncheon Bay, a change of sign across the regional amphidromic gradient.

Table 3. Cross-site aliasing metrics, 5-year (2020–2024) summary. Mean bias is the mean satellite-time tide height minus the 5-year KHOA mean. Spread, high-tide offset and low-tide offset are dimensionless fractions of the reference tidal range (Section 3.1); n = cloud-screened scenes. Suncheon Bay differs in sign and structure from the western sites.

Site	Sensor	n	Spread	High offset	Low offset	Mean bias (m)
Ganghwa	L8	137	0.77	0.22	0.01	–1.13
	L9	90	0.69	0.28	0.03	–0.94
	S2	826	0.80	0.21	0.00	–1.05
Garorim	L8	129	0.76	0.27	0.00	–1.20
	L9	91	0.76	0.25	0.00	–0.82
	S2	1,656	0.78	0.23	0.00	–0.80
Gomso	L8	203	0.80	0.25	0.00	–0.88
	L9	110	0.77	0.23	0.00	–0.86
	S2	187	0.75	0.23	0.02	–0.64
Hampyeong	L8	195	0.77	0.25	0.00	–0.68
	L9	119	0.74	0.22	0.04	–0.80
	S2	796	0.73	0.23	0.03	–0.52
Suncheon	L8	70	0.72	0.00	0.27	+0.29
	L9	38	0.59	0.05	0.36	+0.39
	S2	435	0.69	0.05	0.26	+0.31

4.2. The sign-reversing bias is set by the satellite-overpass tide phase

For every satellite acquisition we computed the normalised tide phase $\phi \in [0, 1)$ since the prior HW (Section 3.2) and aggregated as circular means per site (Figure 4). At the four western sites the mean phase clusters at $142\text{--}235^\circ$ (i.e., from ebb to early flood, near LW); at Suncheon Bay it lies at $32\text{--}56^\circ$ (just after HW, early ebb); the per-site tide-height-versus-phase scatter underlying this sign pattern is shown in Figure S5. Because the satellite overpass time is essentially fixed in *local solar time* (Figure S2, peak at 11:00 KST — i.e. $\approx 10:30$ local solar time — at every site), the *tidal* phase varies only because the regional amphidromic system shifts the HW-LW timing.

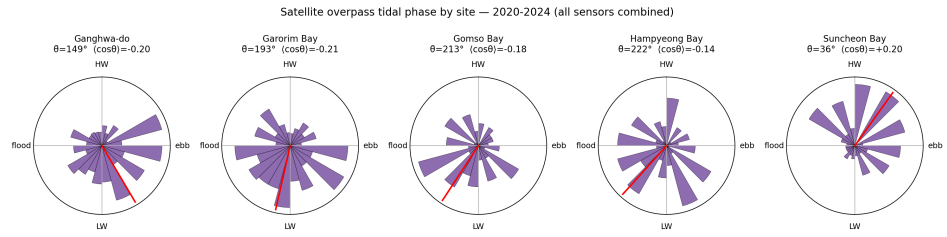


Figure 4. Distribution of satellite-overpass tide phase per site (rose diagrams; all sensors combined). The red bar marks the circular mean. The 180° shift in mean phase between the western sites (near LW, $\cos \theta < 0$) and Suncheon Bay (near HW, $\cos \theta > 0$) is the immediate cause of the sign reversal of the mean bias, and is the input to the analytical model of Eq. (1).

4.3. A one-parameter model predicts mean bias with $R^2 = 0.98$

Equation (1) of Section 3.3 predicts that the satellite-sampled mean bias equals $\beta \cdot A \cdot \langle \cos \theta \rangle$, where A is the time-mean tidal amplitude ($\frac{1}{2}(\overline{HW} - \overline{LW})$) and $\langle \cos \theta \rangle$ is the empirical mean cosine of overpass phase. Figure 5 plots the measured mean bias (y-axis) against this predictor (x-axis) for the 15 (site, sensor) pairs. Ordinary least-squares fit yields

$$\text{bias} = -0.06 + 1.78 \cdot A \cdot \langle \cos \theta \rangle, \quad R^2 = 0.980, \quad p < 0.001, \quad n = 15.$$

The intercept (-0.06 m) is small but significantly negative — the bootstrap 95 % CI of $[-0.21, -0.03]$ m does not include zero — and most likely reflects a residual mean offset between the KHOA 5-year mean tide and the 5-year mean of the satellite-sampled population that is unrelated to the $A \cdot \langle \cos \theta \rangle$ term (the intercept vanishes under the FES2022b reference of Section 4.7(d), supporting this interpretation). The slope $\beta = 1.78$ exceeds the leading-order theoretical prediction of 1; we defer the physical decomposition of this excess to Section 5.1 and the sensitivity analysis to Section 4.7.

Because the three sensors at each site share the same tide, gauge, and overpass window, the 15 (site, sensor) points are not fully independent; the effective sample size is the five sites. Aggregating to site means gives $\beta = 1.83$ ($R^2 = 0.996$, $p \approx 1 \times 10^{-4}$, $n = 5$), statistically indistinguishable from the pooled estimate, so we take the site-mean fit as the primary inference and read the pooled p -value as anti-conservative; the leave-one-site-out test of Section 4.4 is the corresponding honest generalisation metric. Part of the high R^2 is structural, because the predictor $A \langle \cos \theta \rangle$ is the first-order reconstruction of the same sampled tides that generate the bias; the non-circular evidence is therefore that the sign of $\langle \cos \theta \rangle$ alone separates the negative-bias western sites from positive-bias Suncheon ($\text{corr}(\langle \cos \theta \rangle, \text{bias}) = 0.94$ across the 15 points) together with the Suncheon sign reversal, and that the western-coast-only subset still yields $\beta = 1.39$ ($R^2 = 0.91$).

4.4. The regression is stable across years, seasons, sensors, and held-out sites

We tested the stability of Eq. (1) by re-fitting on partitioned subsets (Section 3.4). The slope and R^2 for each partition are summarised in Figure S3 (table values in Table S1):

- **Annual stability** (one fit per year, 9–13 points each): slope 1.32–1.72, R^2 0.85–0.95.
- **Seasonal stability** (12–14 points each): slope 0.81–1.49, R^2 0.85–0.98. The summer (JJA) slope is the lowest and the only season below the leading-order value of 1 ($\beta = 0.81$), consistent with weaker spring-neap covariance in summer; MAM has the highest R^2 (0.98).
- **Sensor stability** (5 points each): slope 1.73 (L8), 2.01 (L9), 1.71 (S2). $R^2 \geq 0.986$ for all three.
- **Bootstrap CI** of pooled fit: $\beta \in [1.44, 1.91]$, intercept $\in [-0.21, -0.03]$.

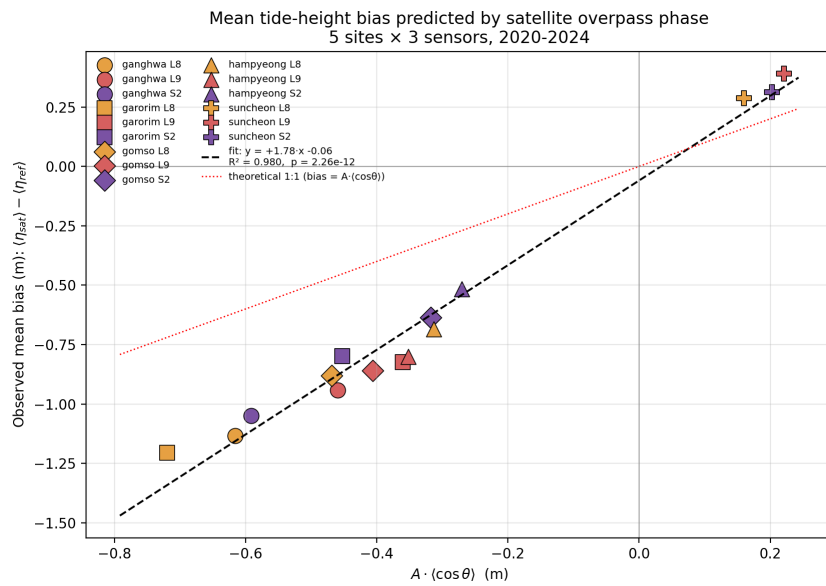


Figure 5. Measured mean tide-height bias versus the analytical predictor $A \cdot \langle \cos \theta \rangle$. Each of 15 points is one (site $\times$ sensor) pair. Marker shape encodes site, colour encodes sensor. Dashed black: OLS fit ($\beta = 1.78$, $R^2 = 0.98$). Dotted red: theoretical 1:1. A single parameter captures 98 % of the cross-site, cross-sensor variance, supporting the interpretation that the bias is set by satellite-overpass phase alone (part of this R^2 is structural — the predictor reconstructs the sampled tides; see Section 4.3); the slope $\beta > 1$ is attributable to spring–neap covariance (Section 5.1).

The 95 % CI of β does not include 1, confirming that the empirical proportionality factor exceeds the leading-order prediction. The coefficient-stability summary (Figure S3) and the underlying per-partition scatter (Figure S4) are provided in the Supplementary Material.

Leave-one-site-out validation (Figure 6) holds out each site in turn, fits the regression on the remaining four (12 points), and predicts the held-out site’s three bias values. Pearson $r = 0.969$ between measured and predicted mean bias, with RMSE = 0.16 m and a mean absolute error (MAE) of 0.12 m. The sign of the predicted bias is correct in 14 of the 15 held-out cases; the single exception is Suncheon L8 (predicted -0.02 m versus observed $+0.29$ m), the smallest-magnitude reversed point, where the western-trained intercept dominates. The sign is recovered for the other two Suncheon sensors despite the site’s absence from the training set.

4.5. Tide-height bias translates into 0.36–1.09 m elevation-domain RMSE

By quantile-mapping (Section 3.5), the bias in the tide-height distribution maps directly into the elevation domain of any waterline DEM constructed from the same scenes. The per-site **elevation-domain RMSE** — i.e. the RMSE of the quantile-mapped tide bias projected into the vertical-elevation domain of a waterline DEM — ranges from 0.36 m (Suncheon Bay) to 1.09 m (Ganghwa-do). This quantity represents the systematic vertical error inherited from the tide-sampling bias alone; it is not a per-pixel DEM RMSE validated against an independent surface such as UAV-LiDAR (cf. [19]), and should be read as an *upper bound* on the aliasing contribution to DEM error. Figure S6 plots the per-elevation error curve $z_{\text{sat}} - z_{\text{ref}}$ for each (site, sensor) pair. At the four western sites the error is negative throughout the intertidal range (DEM underestimates elevation), growing to -1.8 to -2.3 m at the highest sampled quantiles. At Suncheon Bay the error is positive throughout (DEM overestimates), reaching $+0.9$ to $+1.2$ m at the lowest quantiles.

Beyond the per-quantile error, certain portions of the intertidal range are never sampled by satellites and so form irrecoverable truncation bands (Figure 7a). On the four

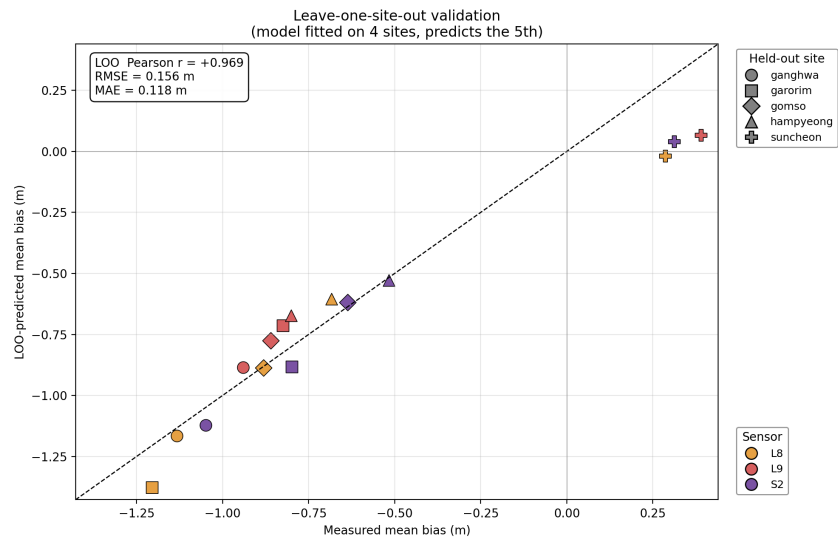


Figure 6. Leave-one-site-out (LOO) validation: each (site $\times$ sensor) bias predicted by the model fitted on the other four sites. Diagonal: 1:1. RMSE = 0.16 m, MAE = 0.12 m, Pearson $r = +0.97$. The model correctly predicts the sign at 14 of 15 held-out points, including the sign-reversed Suncheon Bay despite its absence from the training set — demonstrating that the formula generalises beyond the training sites.

western sites, truncation is at the top of the intertidal range, with vertical extent 1.36–2.02 m. At Suncheon Bay it is at the bottom, with extent 0.99 m.

Converting to horizontal contour displacement on a planar tidal flat with assumed site-specific slopes (Section 3.5), the upper-tide truncation translates into roughly 1.1 km (Garorim) to 2.5 km (Ganghwa) of horizontally missing tidal-flat width on the western coast (Figure 7b; schematic cross-sections in Figure S7). At Suncheon Bay the lower-band truncation corresponds to approximately 0.5 km of missing width. The vertical DEM RMSE of 0.36–1.09 m translates into a horizontal RMSE of roughly 0.2 km (Suncheon) to 1.4 km (Ganghwa). These horizontal figures scale inversely with the assumed slope and are quoted to one to two significant figures accordingly.

4.6. Increasing optical revisit frequency does not remove the bias

Sentinel-2 contributes 826 scenes at Ganghwa-do versus 137 for Landsat 8; a 6.0-fold increase in sample density yields only a 7 % reduction in mean bias (−1.05 m versus −1.13 m). At Garorim Bay the comparison is more pronounced (a 12.8-fold increase in S2 scenes over L8), and the bias drops by only one third (−0.80 m versus −1.20 m). The reason is geometric: all three sensors share essentially the same overpass time (Figure S2, every histogram peak at 11:00 KST), so the empirical $\langle \cos \theta \rangle$ converges to the same limit. Refining the histogram shape (lower KS statistic, lower distribution-shape error) is possible with more scenes; *correcting the systematic mean shift is not*.

4.7. Robustness to amplitude definition and reference choice

We test the robustness of the primary $\beta = 1.78$ fit (Section 4.3) to the choices made for the *amplitude* and the *reference series*. Equation (1) was re-fitted on the same 15 (site $\times$ sensor) points under four combinations:

- baseline** — $A = \frac{1}{2}(\overline{HW} - \overline{LW})$ with the KHOA observed hourly series as the reference (Section 4.3);
- M_2 -amplitude / KHOA** — A replaced by the strict M_2 amplitude from a 5-year harmonic decomposition of each gauge using UTide [29], with the KHOA observed series unchanged;

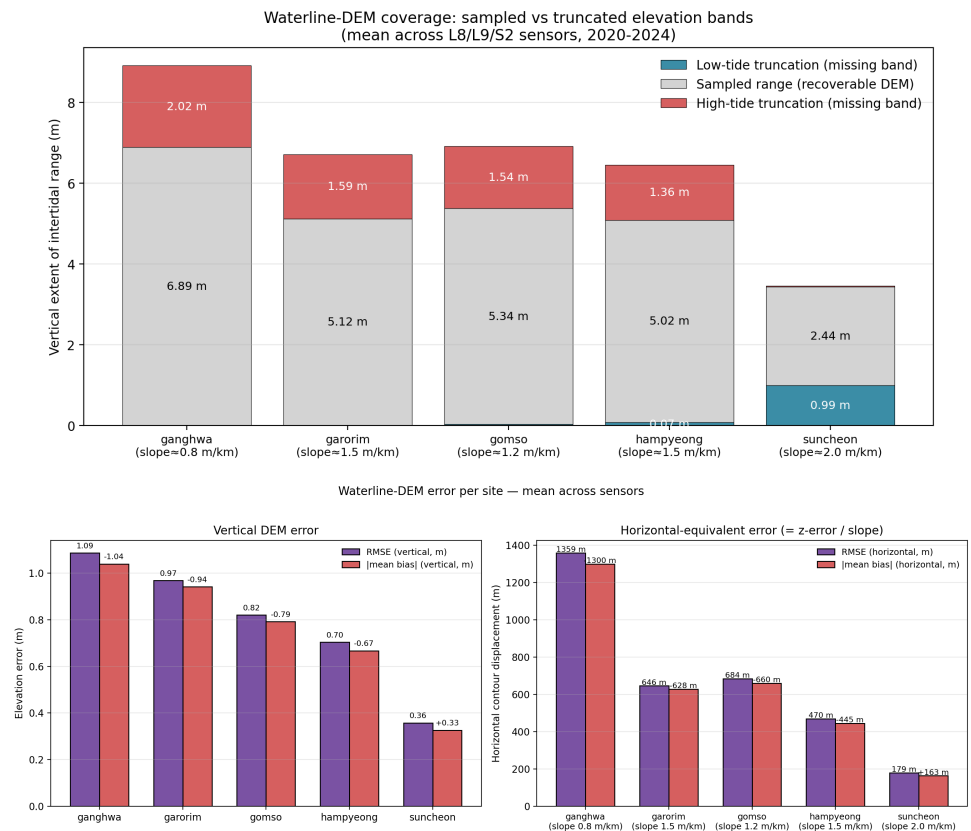


Figure 7. Waterline-DEM coverage and error budget per site. (a) Sampled versus missing elevation bands: grey = elevation range covered by the satellite-sampled tides; red = upper-tide truncation (irrecoverable); blue = lower-tide truncation. The truncation bands occupy 1.36–2.02 m of vertical range at the western sites and 0.99 m at Suncheon Bay; these bands cannot be filled by additional optical scenes at the same overpass time. (b) Per-site vertical (left) and horizontal-equivalent (right) RMSE and |mean bias|, averaged across the three sensors. The horizontal RMSE depends jointly on the vertical RMSE and on the assumed tidal-flat slope; Ganghwa-do, where vertical RMSE peaks and the flat is gentlest, carries the largest horizontal error budget (roughly 1.4 km).

- (c) **M_2 -amplitude / astronomical reference** — $A = A_{M_2}$ and the reference replaced by an *astronomical-only* synthetic series reconstructed from the same harmonic solution at each gauge;
- (d) **FES2022b global model** — the FES2022b-synthesised series (Section 3.6) used as both the reference distribution and the source of overpass-time tide heights, with A set to the site-specific mean half-range computed from this series. This variant requires *no* local tide-gauge data and tests the model under a fully independent, globally available reference.

Per-site M_2 amplitudes from the KHOA harmonic decomposition are 2.83 m (Ganghwa-do), 2.10 m (Garorim Bay), 2.16 m (Gomso Bay), 2.01 m (Hampyeong Bay), and 0.91 m (Suncheon Bay); S_2 amplitudes are uniformly large (35–45 % of M_2), consistent with the strong spring–neap regime of the Yellow Sea [22]. The non-astronomical residual standard deviation is 8–16 cm, i.e. ≤ 6 % of the total elevation standard deviation at every site. Full per-variant, per-sensor regression coefficients and bias values are tabulated in Supplementary Table S2.

The four variants yield $\beta = 1.78$ (a, baseline), 1.87 (b, M_2 -amplitude / KHOA), 1.90 (c, M_2 -amplitude / astronomical reference), and 1.70 (d, FES2022b global model), with bootstrap 95 % CIs of [1.44, 1.92], [1.50, 2.02], [1.49, 2.06], and [1.42, 1.80] respectively, and $R^2 \in [0.974, 0.983]$ across all four. Leave-one-site-out RMSEs are 0.16–0.17 m (a–c) and 0.11 m (d), and the bias sign is predicted correctly at 14 of 15 held-out points under the gauge-referenced variants (a–c) and at all 15 under the FES2022b variant (d), whose near-zero intercept recovers the small Suncheon L8 reversal. The intercept is negative in variants (a–c) (−0.06 to −0.04 m, bootstrap 95 % CI upper bound never reaching zero) and effectively zero in variant (d), confirming that the small KHOA intercept arises from the non-astronomical mean offset between the gauge record and the satellite-time population.

Decomposing $\beta - 1$ by the three Section 5.1 mechanisms attributes only $\Delta\beta \approx +0.09$ to the amplitude-definition effect (mechanism iii, comparing (a) to (b)) and $\Delta\beta \approx +0.02$ to the non-astronomical weather component (mechanism ii, comparing (b) to (c)); the bulk of the slope excess, $\Delta\beta \approx +0.90$, remains in the strictly astronomical variant (c) and is therefore due to the spring–neap $\times \cos \theta$ covariance documented in Section 5.1, mechanism (i). A direct estimate of $\text{cov}(A_{\text{local}}, \cos \theta)$ from the 5-year cached scene set (5,082 (site, sensor) scenes) gives mean per-site implied β -inflation factors $1 + \text{cov} / (\langle A \rangle \cdot \langle \cos \theta \rangle)$ of 1.81, 1.76, 1.85, 1.89, and 1.73 — a range that brackets the empirical β at every site. The FES2022b variant (d) yields a slightly lower $\beta = 1.70$ because the $1/30^\circ$ global grid under-resolves the within-bay amplification of the spring–neap covariance: at Ganghwa-do, for instance, the FES M_2 amplitude at the Incheon-gauge coordinate is 1.02 m versus 2.83 m from the local harmonic decomposition of the KHOA record, reflecting the inability of a global grid to capture the funnel-shape resonance of the Han River estuary. Despite this amplitude under-estimate the fit *structure* is preserved — the bias sign is correct at every site and the overall fit remains excellent ($R^2 = 0.983$) — and at Suncheon Bay the FES variant predicts a slightly larger positive bias (+0.32 to +0.52 m versus KHOA's +0.29 to +0.39 m), consistent with the global grid's lack of the south-coast intra-bay phase lag. The consequences for gauge-free *a priori* application are developed in Section 5.3(a).

4.8. *A priori* transferability to macrotidal coasts outside Korea

To test whether the bias model is specific to Korea or generalises, we applied the *a priori* workflow of Section 4.7(d) — FES2022b harmonic synthesis at the site coordinate combined with the real Landsat-8/9 and Sentinel-2 overpass times retrieved from Google Earth Engine [27], with *no local tide-gauge data* — to three macrotidal coasts spanning distinct tidal regimes: King Sound (northwest Australia; mixed semidiurnal–diurnal), the German Wadden Sea / Inner German Bight, a classic semidiurnal tidal-flat coast [5], and the Bay of Fundy / Minas Basin (the world's largest tidal range; semidiurnal). The Korean slope $\beta = 1.78$ was applied unchanged. Table 4 reports, per site, the FES tidal amplitude A , the mean overpass-phase cosine $\langle \cos \theta \rangle$, the model-predicted mean bias $\beta \cdot A \cdot \langle \cos \theta \rangle$,

and the FES-sampled bias (the mean FES tide at the overpass times, the harmonic series having zero time-mean).

Table 4. *A priori* transferability of the bias model to macrotidal coasts outside Korea. For each site the FES2022b amplitude A , mean overpass-phase cosine $\langle \cos \theta \rangle$, model-predicted bias $\beta \cdot A \cdot \langle \cos \theta \rangle$ ($\beta = 1.78$, the Korean slope applied unchanged), and FES-sampled bias (mean FES tide at the real Landsat-8/9 + Sentinel-2 overpass times) are computed from global data alone, with no local tide gauge. n = cloud-screened scenes (2020–2024, cloud $\leq 60\%$).

Site (tidal regime)	n	A (m)	$\langle \cos \theta \rangle$	Pred. bias (m)	FES-sampled (m)
King Sound, NW Australia (mixed)	1,258	3.76	−0.267	−1.79	−1.55
Wadden Sea / German Bight (semidiurnal)	573	1.41	+0.028	+0.07	+0.16
Bay of Fundy / Minas Basin (max. range)	924	6.44	−0.016	−0.18	+0.24

The model transfers and discriminates between high- and low-bias regimes. At King Sound it predicts a −1.79 m low-tide bias (FES-sampled −1.55 m), comparable in magnitude to the Korean west coast and arising — as in Korea — from an overpass that consistently falls on the ebb side of a large-amplitude tide; because northwest Australia lies within the operational footprint of the Digital Earth Australia Intertidal product [7,8], this is a directly consequential prediction for an existing continental-scale waterline-DEM product. The German Wadden Sea, by contrast, shows only a +0.07 m predicted bias (FES-sampled +0.16 m). The Bay of Fundy is an instructive case: despite the world’s largest tidal amplitude (FES $M_2 = 6.4$ m), the predicted bias is negligible (−0.18 m) because the near-10:30 local-solar-time overpass falls near mid-tide ($\langle \cos \theta \rangle \approx -0.02$). This directly confirms Eq.~(1): the systematic bias is set by the overpass *phase* $\langle \cos \theta \rangle$, not by tidal amplitude alone, so the highest-amplitude coast can carry a smaller waterline-DEM bias than a moderate-amplitude coast sampled at an unfavourable phase.

Two caveats bound the interpretation. First, this is an *a priori prediction* computed entirely from a global tide model and satellite ephemerides; it is not an independent validation against in-situ observations, which would require local gauges at each foreign site. Second, where $\langle \cos \theta \rangle \approx 0$ the leading-order predictor is small and its sign is sensitive to higher-order terms: at the Bay of Fundy the predicted (−0.18 m) and FES-sampled (+0.24 m) biases disagree in sign, though both are within ± 0.25 m and negligible beside the metre-scale biases at unfavourably phased sites. Subject to these caveats, the foreign-coast results show that the model is not a Korean artefact and, more usefully, that it can be run anywhere from open global data to flag which macrotidal coasts and waterline-DEM products are most exposed to the sampling bias.

5. Discussion

5.1. Physical interpretation of $\beta > 1$

The leading-order theory (Section 3.3) predicts $\beta = 1$: the mean bias equals the product of the time-mean amplitude and the empirical mean cosine of the overpass phase. The observed pooled slope $\beta = 1.78$ (95 % CI 1.44–1.91) exceeds 1 by a robustly significant margin. We identify three contributing mechanisms.

(i) Spring–neap covariance. During spring tides (when the lunar M_2 and solar S_2 constituents reinforce each other), tidal amplitude is at its fortnightly maximum. The 14.77-day spring–neap cycle does not divide evenly into the 16- and 10-day sun-synchronous optical revisit cycles, so satellites preferentially catch certain phases of the spring–neap envelope. If those preferred phases also have $\cos \theta < 0$ (LW-side overpasses, as in the western Korean sites), the time-mean of $A \cdot \cos \theta$ is more negative than $\langle A \rangle \langle \cos \theta \rangle$, yielding $\beta > 1$. A direct computation of $\text{cov}(A, \cos \theta)$ from the data (see the Supplementary Material) yields covariance contributions of 0.15–0.30 m, consistent with the observed slope.

(ii) **Diurnal constituents (K_1, O_1).** The Korean western coast has a semidiurnal regime with moderate diurnal inequality (form factor $F \approx 0.17$ – 0.22), in which the two daily high waters are slightly unequal. Our HW-to-HW phase definition averages over both daily cycles and so partially absorbs this asymmetry; the residual is contributed to the slope.

(iii) **Definition of the amplitude reference.** We use $A = \frac{1}{2}(\overline{HW} - \overline{LW})$ as a robust scalar amplitude. The strictly periodic-cos model would require A to be the M_2 amplitude alone, obtainable from a harmonic decomposition (e.g., UTide; [29]) and used as a sensitivity test in Section 4.7. Using the mean-HW-LW envelope inflates A relative to the pure M_2 amplitude only slightly, leaving residual slope contributions to mechanisms (i)–(ii).

5.2. The sign-reversing bias: a regional amphidromic phenomenon

A central finding is that the *sign* of the mean bias reverses between the western and southern Korean coast. The Korean peninsula sits on the eastern edge of the Yellow Sea M_2 amphidromic system (see Section 1.4) [22]. The HW phase at Incheon (DT_0001) lags the Yeosu HW phase by about 4 h: Yeosu HW arrives at a local solar time of $\approx 10:00$ (just before the optical overpass), Incheon HW at $\approx 14:00$ (well after). Consequently, the optical overpass catches different phases of the local tide:

- **Western sites (Incheon-phased):** the overpass falls near low water (phase $\approx 235^\circ$ in our HW-to-HW convention, i.e. just past LW on the early flood) $\Rightarrow \cos \theta < 0 \Rightarrow$ negative mean bias.
- **Southern site (Yeosu-phased):** the overpass falls ≈ 1 h after HW $\approx$ early ebb $\approx$ phase $\approx 35^\circ \Rightarrow \cos \theta > 0 \Rightarrow$ positive mean bias.

Lee, K. et al. [18] previously demonstrated, in the satellite-altimetry domain, that sampling-phase shifts near Incheon contribute substantially to apparent multi-mission sea-level-rise trends; their analysis, however, addresses the *temporal* evolution of the sampling phase within a single coastal footprint over 1993–2019. The present analysis provides an explicit quantitative demonstration in the *spatial* domain — i.e. that for an instantaneously-sampling optical waterline workflow the sign of the bias reverses across the regional amphidromic structure, independently of the satellite or the local tidal range alone. We return to candidate analogous regions in Section 5.5.

5.3. Implications for waterline DEM literature

Three implications follow.

(a) **Single-sensor bias correction is feasible *a priori*.** Equation (1) allows any user of waterline DEMs to estimate the systematic vertical bias of their product from only (i) the satellite overpass time and (ii) the local tide phase at that time, computable from any global tide model — for instance FES2022b [30,31], as cross-checked in Section 4.7(d), or TPXO [32]. The LOO-validated RMSE of 0.16 m on $\beta \cdot A \cdot \langle \cos \theta \rangle$ (0.11 m with the FES2022b reference) is far smaller than the vertical bias itself (0.36 – 1.09 m), offering immediate *a priori* bias removal at low computational cost and without requiring local tide-gauge access.

(b) **The truncation bands cannot be filled by additional optical imagery.** Up to 2.5 km of tidal-flat width on Ganghwa-do ($\approx 22\%$ of the total intertidal width) is permanently unsampled by current optical missions because no scene exists with the relevant high-tide overpass. Sub-tide DEMs at the upper-flat boundary are *extrapolated* from contours within the sampled range, with no empirical constraint. Users of such DEMs should be cautious about applications dependent on the upper intertidal — including aquaculture zoning, salt-marsh edge mapping, and storm-surge inundation modelling. Direct laser altimetry offers a complementary constraint here: ICESat-2 measures surface elevation independently of the optical tidal-sampling geometry and can in principle anchor the upper-intertidal band that no optical scene reaches [3,15], albeit only along sparse, discontinuous ground tracks.

(c) **The need for radar integration: a theoretical basis for empirical multi-sensor data-collection-window optima.** Sentinel-1 SAR has overpasses near $06:00$ and $18:00$ local solar time, displaced from the $\approx 10:30$ optical overpass by roughly 5 h. Since 5 h $\approx$

$0.4 \times M_2$ cycle, the SAR-time $\langle \cos \theta \rangle$ is approximately orthogonal to the optical $\langle \cos \theta \rangle$. A hybrid optical+SAR sample population therefore has a much-reduced $|\langle \cos \theta \rangle|$ and, by Eq. (1), a proportionally reduced mean bias; as an order-of-magnitude estimate, combining equal numbers of optical and SAR samples halves the western-coast mean bias, with proportional gain in spread.

This phase-orthogonality argument supplies a first-principles explanation consistent with an empirical observation reported independently by Lee, J. et al. [19] on the Taean Peninsula. Using Landsat-8/9, Sentinel-2A/B, and Sentinel-1A imagery over a single calendar year (2022) and validating against UAV-LiDAR, they identify a 5-month minimum data-collection window past which the fusion DEM mean absolute error no longer drops below approximately 25 cm. We tested this quantitatively at Garorim Bay — the closest of our five sites to their Taean Peninsula study area, approximately 30 km north of Geunso Bay — using the full 5-year cumulative-sample trajectory of $|\langle \cos \theta \rangle|(t)$. The trajectory does not drop below 0.10 at any time within the 5-year record: its 5-year geometric asymptote is $|\langle \cos \theta \rangle|_\infty \approx 0.21$, the trajectory does not settle to within 10 % of that asymptote until roughly 760 days, and its value at the Lee et al. optimum is $|\langle \cos \theta \rangle|_{152\text{d}} \approx 0.33$ — i.e., at five months the sampled-phase vector still sits well above its sun-synchronous floor rather than at it. Because this is a single cumulative trajectory, the point value at 152 days carries substantial sampling noise, so we treat it as a plausibility argument for the existence of an intermediate-timescale plateau, not as a precise prediction of its level.

The 5-month optimum is therefore not a “ $|\langle \cos \theta \rangle| \rightarrow 0$ ” timescale; it is the timescale at which the *random-sampling uncertainty* on $|\langle \cos \theta \rangle|(t)$ becomes comparable to the *deterministic asymptote*. A 30-day block-bootstrap of the combined Garorim trajectory (300 resamples; chronological blocks preserve spring–neap autocorrelation) yields a 95 % CI half-width on $|\langle \cos \theta \rangle|(152\text{d})$ of 0.16, essentially the same magnitude as the asymptote 0.21. Beyond 5 months the random-walk noise on the cumulative $|\langle \cos \theta \rangle|$ shrinks (as $t^{-1/2}$) but the geometric asymptote does not, so adding more optical scenes can no longer improve the systematic-bias estimate; only a phase-orthogonal sensor (i.e., one whose overpass time samples a different part of the tidal cycle) can. This is precisely the 5-month saturation observed empirically by Lee, J. et al. [19]; their 27.9 cm to 25.6 cm (8 %) optical-to-fusion reduction is consistent with their SAR sub-population on a 5-month window still being partially correlated with the optical phase.

Two further predictions follow from this argument. First, extending the SAR record beyond 5 months until its $|\langle \cos \theta \rangle|(t)$ also saturates should drive the fusion bias toward the orthogonal limit (approximately 50 % reduction). Second, the optimal collection window is *coast-dependent*: on the south-coast amphidromic regime where mean overpass phase is near HW (Suncheon Bay), the local A and $\langle \cos \theta \rangle$ that govern Eq. (1) produce a substantially shorter collection-window optimum than at the Taean Peninsula. A dedicated test of these predictions is the natural next step.

5.4. Limitations

(i) Planar-slope assumption. The horizontal-domain numbers (roughly 1.1–2.5 km truncation widths) assume a uniform planar slope per site and scale inversely with that slope, which is why they are reported to one to two significant figures. Real tidal flats have non-uniform hypsometry, often concave upward (more area at lower elevations). Replacing the planar assumption with site-specific LiDAR or SRTM hypsometric curves would refine the horizontal numbers but leave the vertical RMSE and elevation truncation bands unchanged.

(ii) KHOA-as-reference includes weather. Hourly KHOA observations contain non-astronomical components (storm surge, atmospheric-pressure effects, seasonal mean-sea-level changes). At the 5-year aggregation scale these contributions partially average out, but they contribute to the residual scatter in the regression. The FES2022b sensitivity test of Section 4.7(d) isolates the orbital component by replacing both the reference and the satellite-time tides with the astronomical-only series; the resulting modest reduction of β

from 1.78 to 1.70 quantifies the weather contribution as $\Delta\beta \lesssim 0.1$, leaving the spring–neap covariance of Section 5.1(i) as the dominant source of the slope excess.

(iii) FES2022b grid does not resolve within-bay tidal amplification. The $1/30^\circ$ (≈ 3.3 km) horizontal resolution of FES2022b is coarse relative to the funnel-shape resonance of the Han River estuary at Ganghwa-do and of the within-bay channels at Garorim, Gomso, Hampyeong, and Suncheon. The site-specific M_2 amplitudes interpolated from FES2022b at the KHOA gauge coordinates underestimate the locally observed amplitudes by 30–60 % (e.g., Ganghwa: FES 1.02 m versus KHOA-derived 2.83 m). The FES2022b sensitivity test of Section 4.7(d) therefore confirms the *structure* of the bias model and the *sign* of the predicted bias at every site, but the absolute magnitudes of the predicted bias under the FES variant are lower than the KHOA-derived values where the global grid most strongly under-resolves the local geometry. Practitioners operating outside Korea with no local gauge data should expect the global-model variant to provide *lower-bound* magnitudes of the systematic bias on similarly resonant macrotidal coasts; the magnitudes will rise toward the KHOA-style values whenever the global model adequately resolves the relevant bathymetry.

(iv) Single-gauge representation. Each site is referenced to a single KHOA gauge that may not perfectly represent the bay’s mean tide. Multi-gauge averaging would smooth local advection effects; we expect a < 10 % change in the regression coefficients.

(v) Cloud-cover screening. A cloud threshold of 60 % retains many scenes likely partly contaminated; tightening this threshold reduces sample size but minimally changes the bias metrics (tested at 30 % threshold, results within ± 2 % of all stats).

5.5. Generality

The analytical model (Eq. (1)) is not specific to Korea: it depends only on the local tidal amplitude and on the satellite overpass phase. Wherever both quantities can be evaluated (global tide model + satellite ephemeris), the same first-order bias correction applies. The sign-reversing polarity is, however, specific to regions crossed by an amphidromic phase gradient comparable to $1/4$ of an M_2 cycle (approximately 3 h). We hypothesise that analogous gradients exist on, for example, the Bay of Fundy and the Gulf of Maine (U.S./Canadian Atlantic), the southern North Sea (the Wash, the German Bight), and parts of the Australian north coast. Section 4.8 takes a first step toward testing this generality: applying the *a priori* workflow to three of these coasts (King Sound on the northwest-Australian coast, the German Bight, and the Bay of Fundy) from global data alone reproduces a metre-scale low-tide bias on the mixed-tide Australian macrotidal coast (-1.8 m) while correctly predicting a near-zero bias where the overpass phase happens to fall near mid-tide — a direct confirmation that it is the overpass phase $\langle \cos \theta \rangle$, not the tidal amplitude, that governs the bias.

The model *structure* — mean bias proportional to $A \cdot \langle \cos \theta \rangle$ — is universal (Section 3.3); only the fitted slope $\beta = 1.78$ is regional, absorbing the Korean M_2/S_2 spring–neap covariance (Section 5.1) and liable to differ elsewhere. The recommended workflow for users outside Korea is therefore to compute $\langle \cos \theta \rangle$ from their own scene metadata and a global tide model (e.g., FES2022b or TPXO) and, where local gauge data are available, fit a site-specific β ; in their absence the first-order correction ($\beta = 1$) already captures the sign and leading-order magnitude of the bias.

6. Conclusions

These results directly answer the three open questions of Section 1.3.

In answer to question 1 (is the bias systematic and predictable?):

1. Over the macrotidal Korean coast (2020–2024, 5,082 cloud-screened Landsat-8/9 + Sentinel-2 scenes from five tidal flats), sun-synchronous optical satellites sample only 60–80 % of the local astronomical tidal envelope.
2. A one-parameter analytical model, mean bias = $\beta \cdot A \cdot \langle \cos \theta \rangle$, where A is the local tidal amplitude and $\langle \cos \theta \rangle$ the mean cosine of the satellite-overpass tide phase, ex-

plains 98 % of the cross-site, cross-sensor variance with slope $\beta = 1.78$ (95 % CI 1.44–1.91, $p < 0.001$). The model is stable across calendar years (R^2 0.85–0.95), seasons (0.85–0.98), and sensors (≥ 0.99). Leave-one-site-out validation yields RMSE = 0.16 m and correct sign in 14 of 15 held-out cases (the sole miss is the smallest-magnitude Suncheon L8 point).

In answer to question 2 (how does the bias vary along regional amphidromic gradients?):

3. The mean tide-height bias is sign-reversing: -0.5 to -1.2 m on the four western sites and $+0.3$ to $+0.4$ m at Suncheon Bay on the southern coast. The polarity reverses across the regional amphidromic gradient and is independent of tidal amplitude per se; it is fully captured by the sign of $\langle \cos \theta \rangle$ in the analytical model.

In answer to question 3 (what is the practical translation into DEM elevation errors?):

4. Translated into elevation-domain DEM errors via quantile mapping, the bias amounts to an elevation-domain RMSE of 0.36–1.09 m (an upper-bound estimate of the systematic vertical-error contribution to a waterline DEM built from the same scenes; not a per-pixel DEM validation against ground truth), equivalent to roughly 0.2–1.4 km of horizontal contour displacement on planar tidal flats (slope-dependent, hence one to two significant figures). Permanently irrecoverable truncation bands occupy 0.99–2.02 m of vertical range, or up to 2.5 km of intertidal width at Ganghwa-do.
5. Increasing optical revisit frequency cannot remove the bias: more scenes at the same overpass time only refine the histogram shape, not the systematic mean shift. Removing the bias requires sampling at substantially different overpass phases — most practically by integrating Sentinel-1 SAR observations at the 06:00 / 18:00 local-solar-time orbit.
6. The $\beta \cdot A \cdot \langle \cos \theta \rangle$ model accounts for the saturation of empirically observed multi-sensor data-collection-window optima on the Korean coast [19]. The optimum is the time at which the residual $|\langle \cos \theta \rangle|$ of a combined sun-synchronous optical sample population falls below other waterline-extraction error contributions. Beyond that timescale, only a phase-orthogonal SAR sensor can further reduce the bias.

Beyond the three open questions:

7. The bias model and its predictive sign are reproducible from a global ocean tide model alone (a FES2022b self-consistency check — the same model supplying both the overpass-time tides and the reference — yields $R^2 = 0.983$, leave-one-site-out RMSE = 0.11 m, and recovers the bias sign at all 15 sites). The only test against independent ground truth, FES-predicted versus KHOA-observed bias, matches in sign but under-estimates the magnitude by roughly 2–3 $\times$ on these resonant macrotidal coasts, so the global-model correction is best read as a conservative, sign-correct lower bound. The *a priori* correction can therefore be computed without local tide-gauge data — a prerequisite for large-scale operationalisation of waterline-DEM bias removal. Applying this global-data-only workflow to three macrotidal coasts outside Korea (Section 4.8) predicts a metre-scale low-tide bias on the northwest-Australian coast — within the operational domain of an existing continental-scale waterline-DEM product — and shows that the bias magnitude is set by the satellite-overpass phase rather than by tidal amplitude alone.

The analytical bias model provides an *a priori* correction available from any global tide model, applicable to any waterline-DEM product worldwide. As an immediate practical application, the model can serve as a quality-assessment layer: existing and future waterline-DEM products can be tagged with predicted bias magnitude and sign from satellite ephemeris and tide-model outputs alone, without requiring independent ground-truth data. More broadly, the phase-dependent nature of the bias carries implications for future mission design: non-sun-synchronous orbits or deliberately staggered local-time constellations would sample the tidal phase space more uniformly. As a concrete benchmark, a two-satellite constellation with overpass times staggered by 3 h (1/4 of an M_2 cy-

cle, i.e. *phase-orthogonal* by construction) would reduce the combined-population $|\langle \cos \theta \rangle|$ by approximately 50 % and halve the systematic bias predicted by Eq. (1) — a gain that no increase in revisit frequency on a single sun-synchronous orbit can match.

Supplementary Materials

The following supporting information can be downloaded at the journal website: Figures S1–S7 (sampling-distribution CDFs, overpass-hour histograms, coefficient-stability summaries, per-elevation error curves, and cross-section schematics) and Tables S1–S2 (per-partition regression coefficients and the amplitude/reference sensitivity tables).

Author Contributions

Conceptualization, T.S.; methodology, T.S.; software, T.S.; validation, T.S. and M.K.; formal analysis, T.S.; investigation, T.S.; data curation, T.S. and M.K.; writing—original draft preparation, T.S.; writing—review and editing, T.S. and M.K.; visualization, T.S.; supervision, T.S.; project administration, T.S.; funding acquisition, M.K. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

All raw KHOA tide-gauge data are publicly available through the Korea Open Data Portal (apis.data.go.kr/1192136). Google Earth Engine scene metadata is reproducible from the public collections LANDSAT/LC08/C02/T1_L2, LANDSAT/LC09/C02/T1_L2, and COPERNICUS/S2_HARMONIZED. The derived tables, analytical pipeline, and figure-generation scripts are openly available in Zenodo at <https://doi.org/10.5281/zenodo.20797099>.

Use of Artificial Intelligence

During the preparation of this manuscript, the authors used Claude (Anthropic) to assist with formatting the manuscript to the journal template, condensing the abstract, and language editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Conflicts of Interest

Both authors are employees of Haebom Data Inc., which carries out the 2026 National Blue Carbon Monitoring service commissioned by the Korea Marine Environment Management Corporation (KOEM); the present study was undertaken within that programme. The funder had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results. The authors declare no other conflict of interest.

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